

CO2-AT-1 - Analytical Problem Solving

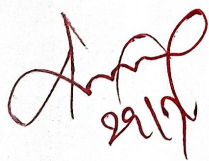
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Course Name: Data Structures

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CO2 - AT - 1

Analytical Problem Solving

- 1) Perform the following operation in LIFO data Structure  
Circular Queue Size = 5, Enqueue(20), Enqueue(10),  
Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90)  
Enqueue(100), Dequeue(), Dequeue(), Enqueue(60) Enqueue(90)

Assume:

Queue Size = 5

Index Position = 0, 1, 2, 3, 4

Initially : Front = Rear = -1

Operation

| Step | operation                | Queue Status       | Front | Rear |
|------|--------------------------|--------------------|-------|------|
| 1    | Enqueue(20)              | [20, -, -, -, -]   | 0     | 0    |
| 2    | Enqueue(10)              | [20, 10, -, -, -]  | 0     | 1    |
| 3    | Enqueue(30)              | [20, 10, 30, -, -] | 1     | 2    |
| 4    | Dequeue() →<br>20 remove | [-, 10, 30, -, -]  | 1     | 2    |
| 5    | Enqueue(80)              | [-, 10, 30, 80, -] | 1     | 3    |
| 6    | Dequeue() →<br>10 remove | [-, -, 30, 80, -]  | 2     | 3    |

|    |                                      |                        |   |   |
|----|--------------------------------------|------------------------|---|---|
| 7  | Enqueue (90)                         | $[-, -, 30, 80, 90]$   | 2 | 4 |
| 8  | Enqueue (100)                        | $[100, -, 30, 80, 90]$ | 2 | 0 |
| 9  | Dequeue() $\rightarrow$ 30<br>remove | $[100, -, -, 80, 90]$  | 3 | 0 |
| 10 | Dequeue() $\rightarrow$<br>80 remove | $[100, -, -, -, 90]$   | 4 | 0 |
| 11 | Enqueue (60)                         | $[100, 60, -, -, 90]$  | 4 | 1 |
| 12 | Enqueue (90)                         | $[100, 60, 90, -, 90]$ | 4 | 2 |

### Final Queue Representation

| Position | 0   | 1  | 2  | 3 | 4  |
|----------|-----|----|----|---|----|
| Element  | 100 | 60 | 90 | - | 90 |

The Queue is traversed circularly

90  $\rightarrow$  100  $\rightarrow$  60  $\rightarrow$  90.



2) Implement Binary Search Algorithm for searching an element and its position having the following 10 elements. Elements are 66, 111, 212, 323, 432, 543, 643, 649, 777, 888. Case 1: Key element = 111, Case 2: Key element = 888, Case 3: Key element = 999.

Given sorted Elements

| Position | 0  | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |  |
|----------|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Element  | 66 | 111 | 212 | 323 | 432 | 543 | 643 | 649 | 777 | 888 |  |

Algorithm:

Set  $Low = 0$ ,  $High = n-1$

2) Compute :

$$Mid = \frac{Low + High}{2}$$

3) Compare Key with  $A[Mid]$

4) If equal  $\rightarrow$  Element found

5) If  $Key < A[Mid] \rightarrow$  Search left half

Time Complexity:

$$O(\log n)$$

Case 1: Key = 111

Iteration 1

$$\text{Low} = 0$$

$$\text{High} = 9$$

$$\text{Mid} = (0 + 9) / 2 = 4$$

$111 < 432 \rightarrow$  Search left half

Iteration 2

$$\text{Low} = 0$$

$$\text{High} = 3$$

$$\text{Mid} = (0 + 3) / 2 = 1$$

Element at  $\text{Mid} = 111$

Case 2: Key = 888

Iteration 1

$$\text{Low} = 0, \text{High} = 9$$

$$\text{Mid} = 4 \rightarrow 432$$

$$888 > 432$$

Iteration 2

$$\text{Low} = 5, \text{High} = 9$$

$$\text{Mid} = 7 \rightarrow 649$$

Search right half

Iteration 3

$$\text{Low} = 8, \text{High} = 9$$

$$\text{Mid} = 8 \rightarrow 777$$

$888 > 777$

Search right half

Result :

- Element Found
- Index Position = 9
- Array Position = 10

Case 3 : Key = 999

Iteration 1

LOW = 0, High = 9

Mid = 4  $\rightarrow$  432

$999 > 432$

Iteration 2

LOW = 5, High = 9

Mid = 7  $\rightarrow$  649

$999 > 649$

Iteration 3

LOW = 8, High = 9

Mid = 8  $\rightarrow$  777

$999 > 777$

Iteration 4

LOW = 9, High = 9

Mid = 9  $\rightarrow$  888

$999 > 888$



LOW = 10, High = 9

Result

→ Element Not Found

### Binary Search Analysis

| Case | Key | Result    | Index Position | Comparisons |
|------|-----|-----------|----------------|-------------|
| 1    | 111 | Found     | 1              | 2           |
| 2    | 888 | Found     | 9              | 4           |
| 3    | 999 | Not found | -              | 4           |

